

Desmond Kofi Boateng

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GitHub — LinkedIn

Education

Boise State University Boise, Idaho
PhD in Computing (Data Science Emphasis) Expected Graduation: Dec. 2028

African Institute for Mathematical Sciences Kigali, Rwanda
MSc in Mathematical Sciences Graduated: Jun. 2024

- Thesis: Comparative Analysis of Singular Value Decomposition Based Methods for Image and Video Compression with Julia.

University of Ghana Accra, Ghana
Bachelor of Arts in Mathematics with Economics Graduated: Nov. 2021

- Thesis: Khovanov's Categorification of the Jones Polynomial.

Technical Skills

- **Languages:** Python, Julia; working knowledge of R and MATLAB
- **Libraries/Frameworks:** PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, requests, BeautifulSoup
- **Tools:** Git, Linux, Bash, LaTeX, AutoHotkey
- **Areas:** Scientific Machine Learning, Operator Learning, Numerical Analysis, Scientific Computing, Partial Differential Equations, Kernel Methods

Work Experience

Graduate Research Assistant Aug. 2025 – Present
Boise State University Boise, ID

- Conduct research on kernel-based methods for operator learning, with a focus on partial differential equations.
- Implemented a more efficient approach for computing the W_2 transform in graph wavelet analysis, reducing computational overhead.
- Conduct numerical experiments to evaluate the accuracy and computational efficiency of operator-learning methods.

Graduate Teaching Assistant Aug. 2024 – May 2025
Boise State University Boise, ID

- Led tutorial sessions and supported student learning in MATH 307: Foundations of Cryptography.
- Updated the course codebase for the first time in nine years, maintaining compatibility with modern standards.
- Developed grading rubrics and graded assignments for MATH 475/575 in Spring 2025.

Machine Learning Engineer
Cape Wesley Consulting

Feb. 2024 – Jul. 2024
Remote

- Built a LegalBERT-based classifier to identify obligations in the U.S. Code of Federal Regulations, achieving 95% accuracy on the evaluation set.
- Developed web scraping and data collection pipelines in Python and AutoHotkey to support complex regulatory and policy-focused research tasks.
- Extracted and processed text from XML-based regulatory documents using Python and regular expressions.
- Contributed to a policy change index linking political events to politicians for downstream analysis.

Tutor for Ghana 1000 Programme
Industry Immersion Africa

Sept. 2023 – Nov. 2023
Accra, Ghana

- Provided tutoring and mentorship to students in the Ghana 1000 Programme, focusing on industry-relevant skills and practical applications.
- Facilitated learning sessions in data analysis and data visualization to strengthen students' employability.

Teaching Assistant
University of Ghana Mathematics Department

Nov. 2021 – Aug. 2022
Accra, Ghana

- Supported introductory calculus and algebra courses for first-year students and organized tutorial sessions.
- Held office hours for students who needed additional academic support.
- Graded interim assessment papers for level 100 and 200 courses.

Selected Projects

Fake News Detection
Python, scikit-learn, XGBoost, NLTK, TextBlob

2024

- Built a machine learning pipeline to classify news articles as fake or real using both news-context and social-network features.
- Trained and compared multiple models, including XGBoost, Gradient Boosting, Logistic Regression, Random Forest, and Decision Tree classifiers.
- Evaluated model performance using 5-fold cross-validation and metrics such as precision, recall, F1 score, accuracy, and AUROC.

Forest Change Detection and Predictive Modeling
Python, pandas, scikit-learn, TensorFlow, statsmodels, matplotlib

2024

- Analyzed tree-cover-loss data to study forest-loss patterns over time and generate forecasts for future loss.
- Implemented predictive models including SARIMA, Random Forest, Gradient Boosting, and LSTM.

- Applied CUSUM-based change-point detection and evaluated model performance using R^2 , mean absolute error, and mean squared error.

Automated Housing Availability Scraper

2024

Python, requests, BeautifulSoup, pandas, APScheduler

- Built a scraper to monitor Boise State University apartment availability and automate updates when units became available.
- Developed an email-notification workflow to send apartment details to specified recipients when availability was detected.
- Automated daily checks using APScheduler, allowing the system to run on a fixed schedule without manual monitoring.

Presentations and Selected Participation

- Selected participant, *Kernel Approximation and Gaussian Processes: Integrating and Expanding Perspectives*, Banff International Research Station (BIRS), Banff, Alberta, Canada, September 13–18, 2026.
- Poster presentation, *Polyharmonic Splines for Operator Learning*, CBMS-AMML 2025, University of Houston, Houston, Texas, December 2025.
- Poster presentation, *Polyharmonic Splines for Operator Learning*, SIAM Pacific Northwest 5th Biennial Meeting, University of Washington, Seattle, Washington, October 2025.
- Talk, *Singular Value Decomposition Based Methods for Image and Video Compression with Julia*, Conference on Mathematics and Its Applications (CMIA 2023), AIMS Ghana, Accra, Ghana, 2023.

Volunteer Experience

Programs Manager

Apr. 2021 – Present

MeetMentors

- Developed the Xplore mentorship program, which has reached about 3,000 university students across Africa.
- Lead the EQUIP program, which has reached about 200 young people across Africa through mentorship in career development, purpose and vision, and leadership development.
- Manage the social media pages of MeetMentors.

Conference Volunteer

Mar. 2026

AMS Spring Western Sectional Meeting

Boise State University, Boise, ID

- Provided volunteer support for the AMS Spring Western Sectional Meeting hosted at Boise State University on March 7–8, 2026.
- Assisted with conference operations and participant support during the meeting.

Volunteer

Nov. 2024, Nov. 2025

Boise State Mathematics Department Math Invitational

Boise, ID

- Volunteered at the Boise State Mathematics Department Math Invitational in 2024 and 2025.

- Supported event coordination and student engagement during the competition.

Volunteer

Allotey Maths Camp 2023

Sept. 16, 2023 – Sept. 23, 2023
Achimota High School, Accra, Ghana

- Developed content on cryptography for high school students.
- Worked through daily puzzles with students.
- Facilitated lab sessions for different class sections.

Awards and Honors

- Conference Travel Grant (\$500), SIAM Pacific Northwest 5th Biennial Meeting, University of Washington, Seattle, Washington, October 2025.
- Conference Travel Grant (\$900), CBMS-AMML 2025, University of Houston, Houston, Texas, December 2025.

Leadership and Service

President

International Student Association

2025 – 2026
Boise State University

- Provide leadership for the International Student Association and support programming for the international student community.
- Work with student leaders and university stakeholders to help plan events and initiatives that strengthen student engagement.

Graduate College Senator

Associated Students of Boise State University (ASBSU)

2025 – 2026
Boise State University

- Represent graduate student interests in student government discussions and campus initiatives.
- Contribute to efforts related to graduate student advocacy and university engagement.

Member

Graduate Student Advisory Council on Mentoring and Advising

2025 – 2026
Boise State University

- Serve on a council focused on mentoring and advising issues affecting graduate students.
- Contribute to discussions and recommendations aimed at improving graduate student support.